Protocol: 75
Human Self-Discipline Credit Protocol

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Abstract

This whitepaper outlines a decentralized behavioral contract and on-chain reputation protocol built on Move smart contracts—Protocol: 75. Addressing the dual structural dilemma of the universal difficulty in maintaining personal self-discipline and the low efficiency of the traffic economy in traditional centralized platforms, this protocol proposes a "distributed compliance game mechanism based on strong social relationships".

Within the protocol network, users form decentralized consensus units based on a Prior Trust Graph, locking up crypto assets as an irrevocable commitment to self-discipline. Smart contracts act as impartial adjudicators, automatically executing the redistribution of default penalties based on on-chain compliance results, thereby transforming originally loose social pressure into an automated economic game with strong binding force; meanwhile, the underlying assets are automatically routed to verified DeFi protocols, earning risk-free Real Yield for users.

Building upon this, the protocol constructs a "Bio-Credit" system. This credit score, acting as a non-transferable Soulbound Token (SBT), strictly exhibits logical growth corresponding only to the user's authentic compliance behavior. It serves as the value hub connecting individual C-end consumers with the B-end commercial ecosystem: brands can directly inject incentive funds into high-credit-score groups, replacing traditional ad-buying, and forging a disintermediated commercial closed loop where "behavior equals value".

In terms of the economic model, this protocol strictly adheres to the Anti-Ponzi principle, establishing a closed economic system with transparent funding sources that is completely independent of the liquidity from late entrants. All value flows within the protocol originate solely from three channels: the redistribution of defaulters' penalties, the financial interest from underlying assets, and the commercial capital injection from B-end brands. Concurrently, the protocol abides by an anti-gambling baseline; through the supervisory mechanism of strong social relationships and a strict single-staking limit, economic incentives are rigorously confined to a reasonable range that aids self-discipline, completely stripping away any speculative attributes.

On the technical foundation, the protocol utilizes the unique Resource-Oriented features of the Move language to implement strictly non-custodial fund management. All staked assets are decentrally stored as independent Static Resource Objects within the users' respective on-chain account spaces, never congregating in any centralized Liquidity Pool. This architecture converges the hacker attack surface at the root, eliminates the risks of stolen liquidity pools or Rug Pulls common in traditional DeFi protocols, and ensures absolute sovereignty and long-term robust operation of protocol assets.

1 Introduction

1.1 The Personal Dilemma: Combating Entropy and the Assetization of Willpower

In behavioral economics, "Time Inconsistency" reveals a cruel human weakness: in the absence of strong external constraints, short-term temptations often easily defeat long-term goals. Relying solely on internal drive to maintain long-term self-discipline is almost contrary to biological instincts. In our traditional digital lives, the sweat and effort we pour into the gym or on the court are often isolated and fleeting, ultimately reducing to useless "data dust" that cannot be captured or precipitated into any lasting value.

The birth of this protocol is designed precisely to break this dilemma. Through smart contracts, we have constructed a rigid compliance mechanism that impartially adjudicates and transforms the user's fleeting willpower, along with every victory against inertia, into an immutable on-chain asset—"Bio-Credit". Unlike traditional credit systems that rely on financial lending data, Bio-Credit is a sovereign-level reputation completely independent of any centralized platform. It is not pegged to asset scale, nor does it rely on a platform's charity; it grows solely with the user's authentic efforts. When this credit accumulates to a specific height, it will become the hard currency for users to acquire high-tier privileges and commercial returns within the Web3 world.

1.2 The Commercial Revolution: Crushing Intermediaries and the End-to-End Reconstruction of Value

Meanwhile, a massive number of commercial brands (especially in the pan-health, insurance, and AI sectors) are deeply bogged down in the quagmire of the traffic economy. The current commercial advertising model is essentially algorithmic exploitation: brands are not only forced to bear extremely high Customer Acquisition Costs (CAC), but they also face traffic fraud and extremely low conversion rates brought about by algorithmic black boxes. Huge marketing budgets are worn away by intermediary platforms through layers of funnels, leaving only minuscule funds that actually reach target users. Yet, the true value creators—highly disciplined

users possessing extremely high Life-Time Value (LTV)—are reduced to mere "traffic packages" peddled by platforms, unable to obtain their rightful returns.

This protocol aims to reconstruct this inefficient rent-seeking model from the ground up. Using the Bio-Credit system as our entry point, we establish a frictionless "direct attention channel" for B-end brands. Brands no longer need to purchase massive amounts of expensive, ineffective exposure from centralized platforms; instead, they can directly inject their advertising budgets—which would otherwise be exploited through layers of the marketing funnel—into the protocol's incentive prize pool. Through transparent filtering via on-chain data, brands can accurately lock onto authentic users holding high Bio-Credit scores and distribute funds directly to them.

This is no longer advertising placement; this is a direct value handshake between brands and core users. We are returning capital efficiency to its absolute extreme, ensuring that every cent of the budget flows to genuine self-discipliners rather than traffic intermediaries. This fundamentally bypasses the attrition problems of traditional marketing funnels and constructs a new type of production relations based on real behaviors and direct incentives.

2 Core Mechanisms of the Protocol

2.1 Account and Identity System

- **Permissionless Access**: Built on top of the Aptos public chain, any on-chain address compatible with standard wallet protocols can access the protocol.
- **Account Admission**: When creating an account for the first time, a user needs to pay 0.75 USD (paid in protocol-supported stablecoins) as a one-time protocol entry fee.
- **Bio-Credit**: Upon initial access, a Soulbound Token (SBT) is automatically minted as the sole carrier of Bio-Credit, with an initial score of 50. It is non-transferable, indestructible, and its state can only be modified through compliance behaviors.
- **Aggregated Assets**: The Bio-Credit SBT follows the Move language's extensible object standard, supporting the mounting of sub-resources (e.g., commercial medals issued by the B-end, winning streak achievement badges), forming a growable on-chain reputation container.
- **Physical Binding**: The SBT forcibly requires a one-to-one hash mapping with the user's physical health account (such as Apple Health, Google Fit). The protocol verifies hardware fingerprints through an oracle.

2.2 Task System

- **Task Atom**: The protocol defines a set of standardized, extensible "Task Atom" objects. The current version supports three types: daily exercise, compliant sleep, and mindfulness meditation.

- **Task Combo**: Users can freely combine multiple Task Atoms and set specific parameters (e.g., "burn 500 kcal daily + meditate for 10 minutes") to generate independent Task Combos.
- **Difficulty Synthesis**: Based on the parameter intensity of the Task Combo, the protocol contract synthesizes the challenge's "Difficulty" coefficient through preset algorithms to calculate the "Daily Energy Min".
- **Verification Standard**: The protocol adopts an "off-chain computation, on-chain verification" model. After an Oracle equipped with anti-cheating algorithms fetches and verifies the signature data of the user's actually bound physical hardware, it only puts the generated Proof Hash and the result boolean value on-chain.

2.3 Bio-Credit System

- **Credit Intervals**: The three intervals are the Repair Phase (35~50 points), Accumulation Phase (50~75 points), and Elite Phase (75~95 points).
- **Compliance Judgment**: At the start of a challenge, a target number of days is set. If the number of days the target is met is not less than this target, it is considered compliance; otherwise, it is a default.
- **Daily Check-in**: Users need to check in manually every day. The system automatically fetches physical hardware data, performs anti-cheat signature verification, and calculates whether the daily target is met.
- **Energy**: After daily check-in, regardless of whether the target is met, an energy value is calculated based on today's completion status. After the challenge ends:
  - **Valid Energy**: When a user successfully complies, the total energy accumulated during their challenge cycle is considered valid and used to calculate the change in credit score.
  - **Invalid Energy**: If a user is deemed to have defaulted, all energy accumulated during their challenge cycle is forced to zero. The system will execute deduction penalties on the existing credit score based on the number of non-compliant days.
- **Resistance Cost**: Energy changes are influenced by a piecewise linear resistance function. The higher the current credit score, the higher the resistance cost to acquire energy.
- **Energy Decay**: Simulating biological metabolic laws (System Entropy), the protocol deducts a fixed basic value from the current energy slot every day, stopping when the energy reaches 0.

2.4 Challenge and Settlement

- **Asset Standard**: The protocol exclusively uses USD stablecoins as the sole staking and settlement asset across the board to eliminate the interference of crypto asset volatility on behavioral incentives.

- **Challenge Initialization**:
  - **Distributed Hash Teaming**: Regular users must form a consensus group of 4 to 7 people. The protocol contract generates a unique Challenge Hash as an identity index based on each team member's account address and the ID of the currently challenged Task Combo.
  - **Whitelist Mode**: Certified B-end merchant addresses can initiate challenges with no limit on the number of participants (e.g., a 10,000-person marathon), usually accompanied by commercial capital injection.
- **Challenge Cycle**: Must be a positive integer multiple of a week (604,800 seconds) to reinforce periodic habit formation.
- **Fund Redistribution**:
  - **Full Compliance**: The principal is fully refunded to respective accounts, and the underlying financial yield is divided equally.
  - **Partial Default**: Triggers the internal compensation mechanism. The entire principal and interest of defaulting members will be directly split equally among the successfully complying teammates in the group after deducting the liquidation service fee (see 2.7).
  - **Full Default Circuit Breaker**: Triggers the black hole mechanism. All principal and interest will be fully confiscated by the protocol (injected into the treasury and public welfare), and no one receives a refund.
- **Credit Isolation**: Joint liability at the fund level does not affect the credit level. Changes in the credit score depend solely on the user's personal compliance status; teammates' defaults will not cause complying users to lose points.

2.5 Non-Custodial Asset Management

- **Underlying Yield-Bearing**: During the challenge cycle, staked assets will be automatically routed to audited on-chain principal-protected financial protocols (such as Aries Market, Thala) to obtain risk-free yields.
- **Non-Custodial Ownership**: After the original funds enter DeFi protocols, the yield-bearing tokens (LP Tokens) are not pooled in a central contract; instead, they are returned and stored in the Move resource module under the user's independent account. Users have sole ownership of the asset, and the protocol only has management authority, not misappropriation authority.
- **Escape Pod Mechanism**: If the off-chain settlement oracle fails to respond overdue (T+7 days), the protocol automatically triggers a safe mode and opens a mandatory withdrawal interface. Users can unconditionally retrieve their principal using the yield-bearing tokens in their accounts, eliminating the risk of a single point of failure.

2.6 Rights and Commercial Value

- **Rights Threshold**: Users with a Bio-Credit score greater than or equal to 75 automatically obtain whitelist permissions to participate in B-end merchant-funded tasks, translating attention and sweat into direct economic returns.
- **Data Contribution**: With user authorization, verifiable compliance data will serve as Real World Evidence (RWE) to construct human behavioral datasets, providing data sources for data industries such as AI training and insurance actuarial science. Data revenues will be fed back to personal accounts.

2.7 Protocol Economic Model

This protocol constructs a transparent, sustainable economic closed loop with positive externalities. Protocol revenue will be strictly used to maintain system security, cover infrastructure costs (such as oracles and storage), and support on-chain public welfare causes.

- **Entry and Minting Tax**: Of the fee paid by users upon initial access, 80% flows into the protocol treasury, and 20% is donated to on-chain public welfare addresses.
- **Yield-Bearing Service Fee**: Of the yields generated by underlying yield-bearing, 90% goes to the users, 8% flows into the protocol treasury, and 2% is donated to on-chain public welfare addresses.
- **Liquidation Service Fee**:
  - **When partial default occurs**: 95% of the default penalty is equally divided among other complying teammates, 4% is confiscated by the protocol treasury, and 1% is donated to on-chain public welfare addresses.
  - **When a full circuit breaker occurs**: 80% of the default penalty is confiscated by the protocol treasury, and 20% is donated to on-chain public welfare addresses.
- **B-End Commercial Tax**: When external institutions establish commercial prize pools on the protocol, they must pay 10% of the total prize pool amount as protocol revenue. Of this amount, 8% flows into the protocol treasury, 2% is donated to on-chain public welfare addresses, and the remaining 90% serves in full as rewards for high-credit users.

3 Credit Algorithm Model

The Bio-Credit algorithm model aims to establish a fair, anti-inflationary reputation measurement standard with physical significance. Unlike the compound interest growth of financial assets, the growth of Bio-Credit follows a "Biological Resistance Model"; as the score increases, the difficulty of maintaining and breaking through it should rise non-linearly. This chapter details the transformation logic from physical behavior to on-chain credit.

3.1 Core Parameters and Definitions

The anchoring baseline for the following parameter settings is: a medium-intensity, single-task user (defined as: completing only the "Daily Exercise" task daily, with an intensity of 500kcal + 60min), whose theoretical cycle to transition from the Accumulation Phase (50 points) to the Elite Phase (75 points) is designed to be 168 days (considering real-life friction, practically about 180 days). For the specific calculation process, refer to Appendix B.

Constant Name	Set Value	Description
DECIMALS	1,000,000	Precision, e.g., storing 50,000,000 on-chain represents a credit score of 50.0.
ENERGY_SCALING_FACTOR	500	Energy scaling factor. This coefficient is used to convert physical indicators into on-chain energy units, and is the core constant for balancing total energy demand with daily output capacity.
ENERGY_DAILY_ENTROPY	25,000	Daily energy system entropy. This value simulates biological metabolic loss, representing the minimum energy cost required for a user to maintain their current credit score every day.

The current protocol defines three standard Task Atoms, and their respective per-unit difficulty weights W are initialized as follows:

No.	Task Atom	Task Indicator	Unit	Per-unit Difficulty Weight	Optional Target Lower Bound	Optional Target Upper Bound
1	Daily Exercise	Calorie Burn	Kilocalorie: kcal	$W_{kcal} = 1$	200	2000
		Duration	Minute: min	$W_{min} = 3$	30（0.5h）	300（5h）
2	Compliant Sleep	Target Sleep Time	Second：s	0	0	86399
		Target Wake Time	Second：s	0	0	86399
		Valid Duration	Quarter-hour：quarter	$W_{quarter} = 5$	28（7h）	36（9h）

3	Mindfulness Meditation	Duration	Minute：min	$W_{min} = 5$	10	120（2h）

3.2 Energy Production

Every compliance behavior of the user will first be converted into "Energy", which acts as the intermediate medium connecting the physical world and digital credit. In this phase, we initially calculate only the gross energy produced, without deducting entropy consumption just yet.

3.2.1 Gross Energy Production

If the energy scaling factor is denoted as E_{sf} , and the actual completed value of a task on a given day is denoted as V , then the gross energy obtained on that day is as follows:

$$E_{gross} = E_{sf} \times \sum (V \times W)$$

Similarly, if the task goal is denoted as G , the minimum gross energy that should be earned daily is calculated as follows, which can be used for challenge default settlement:

$$E_{goal} = E_{sf} \times \sum (G \times W)$$

Scenario A: a relatively standard user model

- Goal: 500 kcal + 60 min.
- Base Difficulty: $500 + (60 \times 3) = 680$.
- Gross Output: $680 \times 500 = 340,000$.

Scenario B: a highly disciplined user model

- Goal: Standard exercise + 8 hours of compliant sleep + 20 minutes of mindfulness meditation.
- Total Difficulty: $680 + 160 + 100 = 940$.
- Gross Output: $940 \times 500 = 470,000$.

3.2.2 System Entropy and Net Energy

To comply with the Second Law of Thermodynamics, or the principle of "sailing against the current; if you do not advance, you retreat", the protocol introduces the "Daily Energy System

Entropy" mechanism. Regardless of the user's performance on that day, a basal metabolic energy of E_{de} points for the daily energy system entropy must be deducted.

If a standard user model for a period of t weeks completes d days, where $i \in [1, 2, ..., 7t]$, then the total net energy obtained during the challenge is:

$$E_{net} = \max(0, \sum_{i=1}^{d} E_{gross}^{i} - (7t * E_{de}))$$

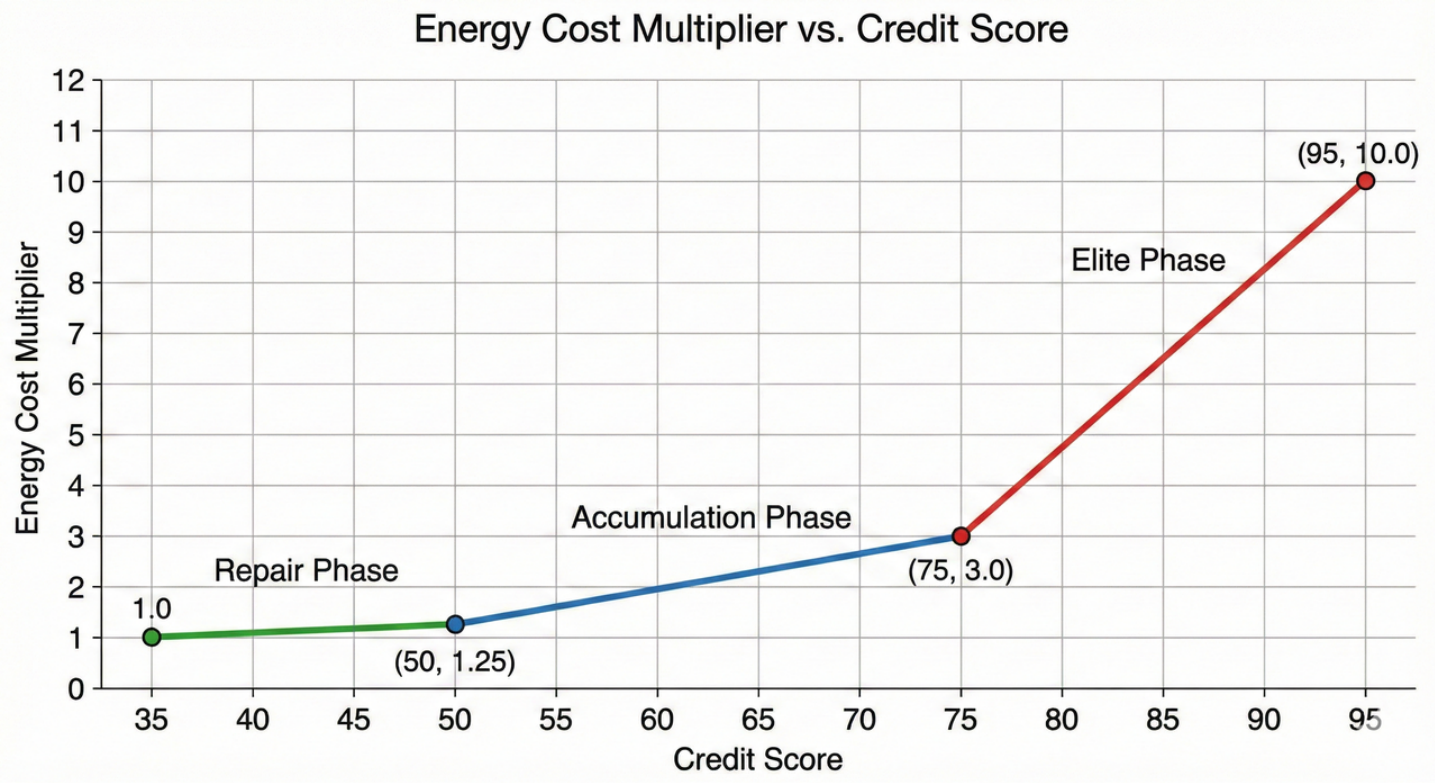
Please also note: This mechanism remains in effect during the rest periods between challenges. Unless it is the user's first challenge, it is settled first when the user starts a new challenge.

For example, a standard user model for a one-week challenge rested for 3 days before this challenge and completed 5 days during the challenge. The total net energy obtained would be: $5 \times 340,000 - (7 + 3) \times 25,000 = 800,000$.

3.3 Credit Calculation

3.3.1 Three-Phase Resistance Model

As the score increases, the "Cost" (resistance) to raise the score grows in a piecewise linear manner. The resistance multiplier $C(s)$ is defined as: the net energy consumed (unit: millions) required to increase the score by 1 point. We have three different credit intervals; although a non-linear relationship should be used in principle, considering the simplification of on-chain calculations to save Gas, the functional relationship between the energy cost multiplier and the current credit score is designed as a three-segment linear function, as shown in the figure and table below:



Energy Cost Multiplier vs. Credit Score

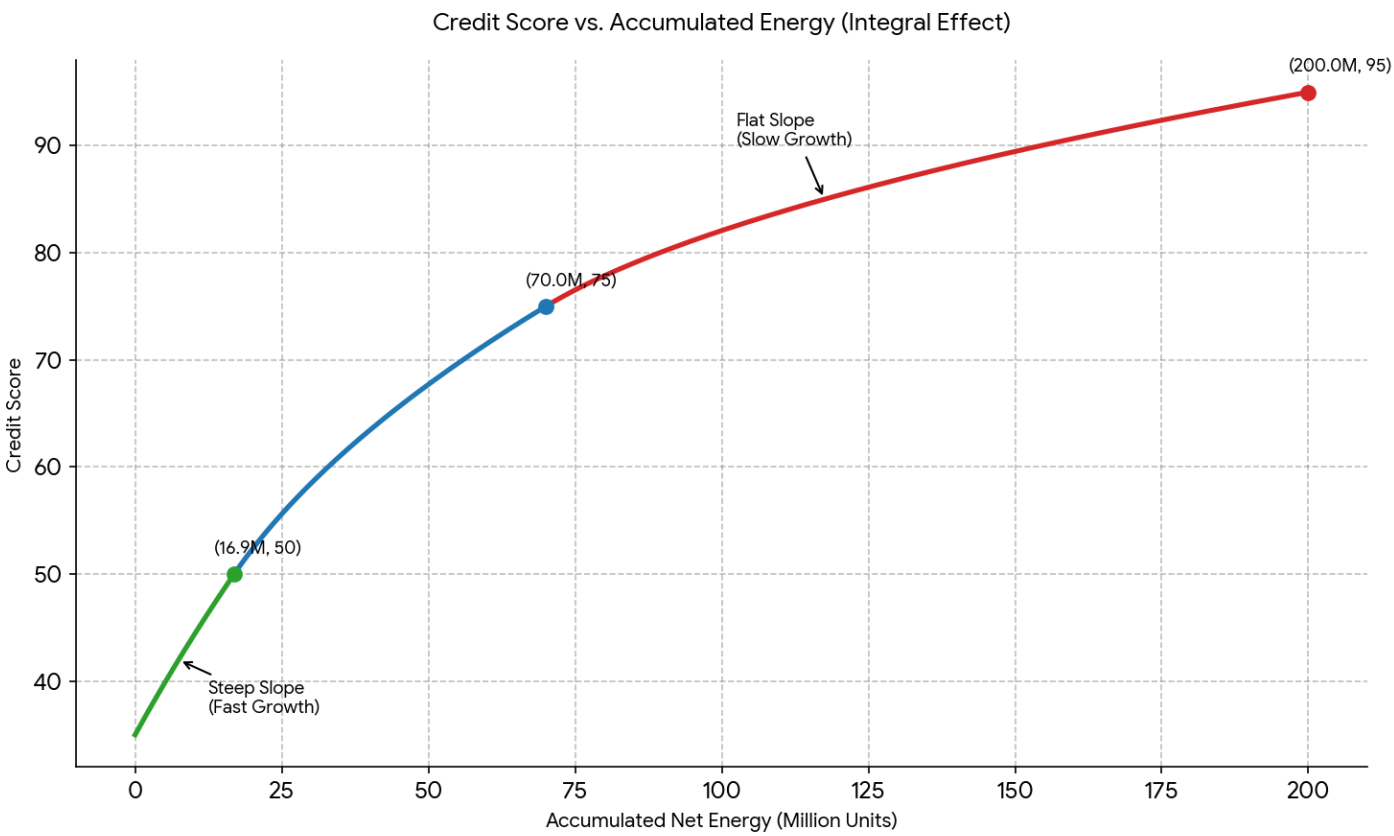
Credit Interval	Score Range	Energy Cost Multiplier	Description
Repair Phase	35 ~ 50	1.0 ~ 1.25	Aimed at rapidly repairing credit, restricting certain privileges.
Accumulation Phase	50 ~ 75	1.25 ~ 3.0	Gentle score increase, objectively reflecting the formation of compliance habits.
Elite Phase	75 ~ 95	3.0 ~ 10.0	Difficult to increase score, default penalties multiplied, extremely hard to reach a perfect score.

3.3.2 Energy Calculus to Credit

This protocol adopts a real-time differential and integral algorithm to convert the total net energy E_{net} into the corresponding credit score. Whether adding or deducting points, or regardless of how much energy has been hoarded, the score change corresponding to each unit of energy depends on the resistance multiplier $C(S_t)$ at the current moment.

$$S_{new} = S_{old} \pm \int_0^{E_{net}} \frac{1}{C(S_t)} dE$$

This effectively prevents users from hoarding massive amounts of energy in the lower score brackets and then using it all at once to leap through the high score brackets (exploiting low prices for arbitrage). For example, if a user submits a massive amount of energy at 49.9 points ($Cost \approx 1.25$), the system will first consume a minimal amount of energy to push the credit score to 50.0; all remaining energy must forcibly be converted at the higher resistance ($Cost > 1.25$) past 50.0.



Credit Score vs. Accumulated Energy (Integral Effect)

- On the left side, the curve is very steep. This means that only a little energy is needed for the credit score to rise quickly.
- Moving to the right, the curve becomes increasingly flat, which means that despite expending a large amount of energy, the credit score can only increase slightly.

3.3.3 Numerical Approximation Implementation of On-Chain Calculus

Although the continuous integral model defined in Section 3.3.2 is mathematically extremely elegant, directly solving this calculus equation in a real blockchain execution environment faces severe engineering challenges. To adapt to the characteristics of blockchain virtual machines and reduce Gas costs, the protocol adopts scaled integers for all calculations across the board; that is, all scores and energy are stored on-chain as u64 integers scaled up by 10^6 times, referring to the precision constants in Section 3.1.

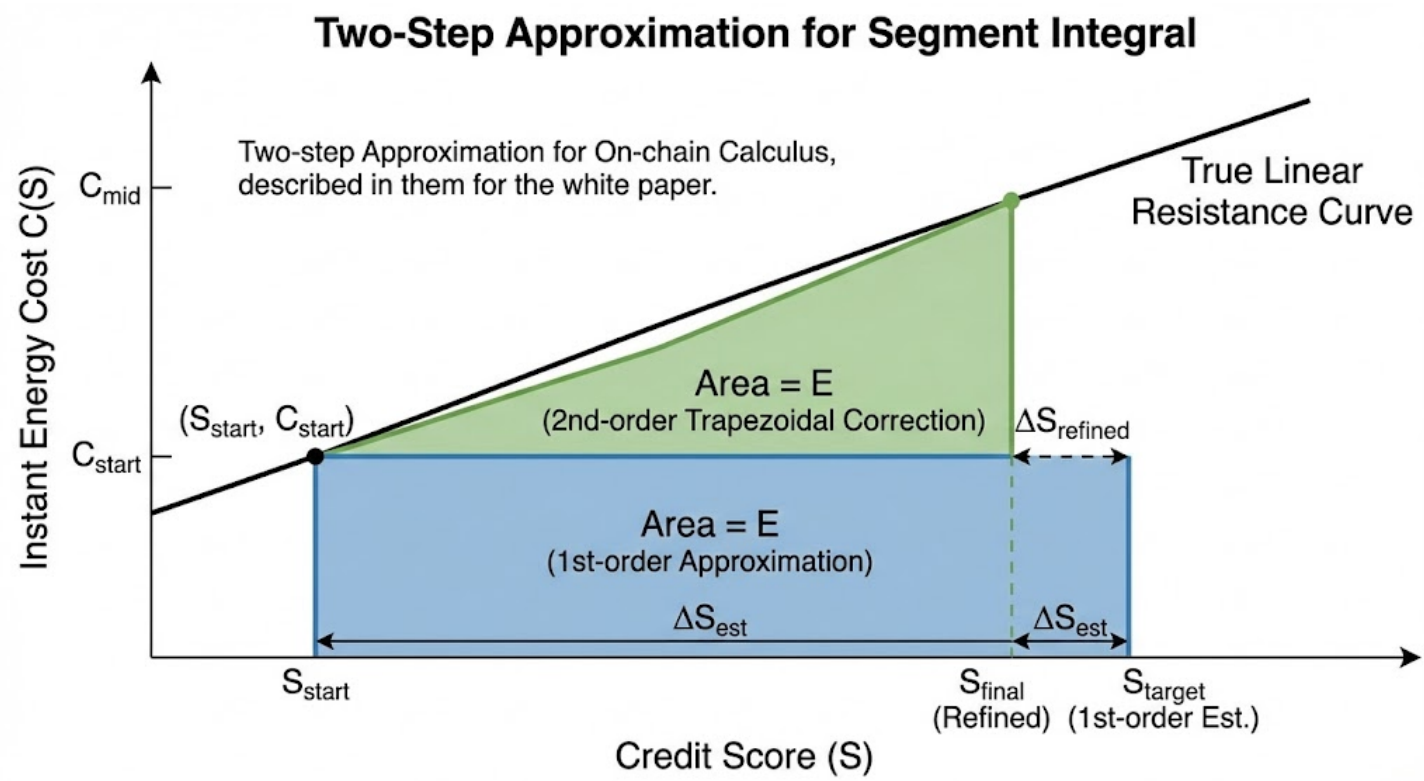
Mathematically, since the energy cost multiplier $C(S)$ is piecewise linear, if we want to find the exact increase in credit score ΔS given the consumed energy E within a certain interval, its

essence is solving for the base width of a trapezoid. Its corresponding algebraic equation is a quadratic equation:

$$E = \frac{1}{2}k(\Delta S)^2 + C_{start}\Delta S$$

To solve for ΔS, it is inevitable to use the root formula, which will introduce the square root ($\sqrt{x}$) operation. In a smart contract environment like the Move language, which lacks native floating-point support and where computational resources are extremely expensive, executing high-precision integer square roots not only excessively consumes Gas but also triggers complex precision loss and overflow risks.

Therefore, this protocol abandons inefficient exact analytical solutions and designs a two-step approximation method specifically optimized for on-chain execution. This algorithm relies solely on basic addition, subtraction, multiplication, and division operations; while ensuring an extremely low computational complexity of $O(1)$, it achieves a precision infinitely approaching the true integral.



Two-Step Approximation for Segment Integral

A. Forward Calculus (Compliance Score Addition)

When a user successfully complies and obtains available energy E_{total}, the contract will attempt to cross the resistance intervals segment by segment in order from low score to high score (Repair Phase $\rightarrow$ Accumulation Phase $\rightarrow$ Elite Phase). The calculation steps within a single interval are as follows:

1. **Cross-segment Detection:**
   a. The contract first uses the trapezoidal area formula to trial-calculate the total energy E_{full} required if it runs fully through the current entire interval (from S_{start} to the upper limit of this segment S_{max}).

$$E_{full} = \frac{C_{start} + C_{end}}{2} \times (S_{max} - S_{start})$$

   b. If $E_{total} \geq E_{full}$, it indicates sufficient energy, directly pushing the current score to the upper limit S_{max} of that segment; after deducting E_{full} , it carries the remaining energy into the next higher resistance interval to continue calculations.
2. **Intra-segment Approximation:** If the energy is insufficient to finish the current segment interval, it indicates that the target landing point inevitably falls inside the interval. At this time, the two-step approximation algorithm is triggered:
   a. **First-order approximation:** Assume that in the upcoming minor sliding, the resistance remains constant (equal to the starting point's resistance C_{start}). Use this to roughly estimate a temporary step length and a temporary landing point S_{target} :

$$\Delta S_{est} = \frac{E_{total} \times dcms}{C_{start}}$$

$$S_{target} = \min(S_{start} + \Delta S_{est}, S_{max})$$

   b. **Second-order correction:** Based on the temporary landing point S_{target} obtained from the first-order approximation, use linear interpolation to calculate the true instantaneous resistance C_{mid} at that point. Subsequently, take the average of the starting point's resistance and the midpoint's resistance as the equivalent resistance $\overline{C}_{est}$ for this path, and use this to recalculate the exact final moving distance $\Delta S_{refined}$:

$$\overline{C}_{est} = \frac{C_{start} + C_{mid}}{2}$$

$$\Delta S_{refined} = \frac{E_{total} \times dcms}{\overline{C}_{est}}$$

3. Finally, derive the exact score after the point addition:

$$S_{final} = \min(S_{start} + \Delta S_{refined}, S_{max})$$

B. Reverse Calculus (Default Penalty and System Entropy Consumption)

When triggering default penalties or settling the system entropy debt, the protocol needs to execute a reverse integral for deducting energy. Its algorithmic core is similar to the forward calculus, but strictly in the opposite direction: the contract will drop back segment by segment from the user's current high-score interval to the low-score interval (Elite Phase $\rightarrow$ Accumulation Phase $\rightarrow$ Repair Phase).

The reverse calculation also follows the two-step rule during intra-segment approximation:

1. **First-order approximation (Downward addressing):** Use the resistance C_{start} at the current high-score point as the baseline to estimate the temporary downward drop distance ΔS_{est} . Since points are being deducted, the temporary landing point is:

$$S_{target} = \max(S_{start} - \Delta S_{est}, S_{min})$$

2. **Second-order correction (Downward smoothing):** Look up the lower resistance C_{mid} at the temporary landing point. Since the lower the score, the smaller the resistance, the actual drop distance will be deeper than the first-order approximation. The contract calculates the true drop distance through the average equivalent resistance $\overline{C}_{est}$:

$$\Delta S_{refined} = \frac{E_{total} \times decimals}{\overline{C}_{est}}$$

3. Finally, derive the exact score after the point deduction:

$$S_{final} = \max(S_{start} - \Delta S_{refined}, S_{min})$$

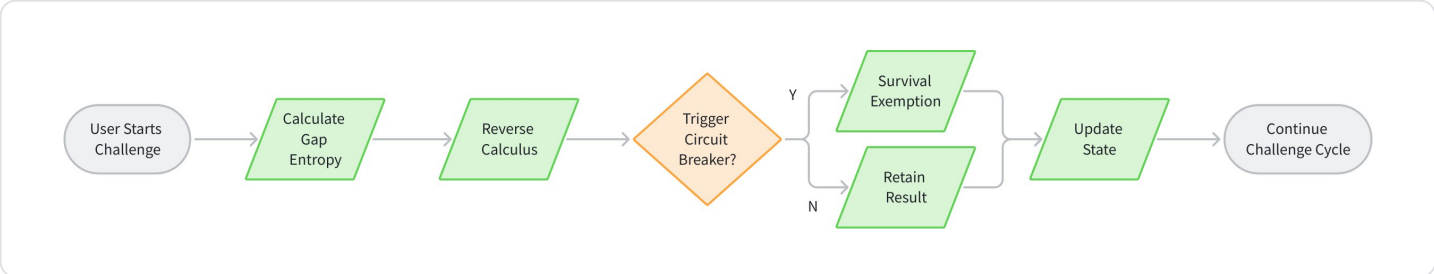
This algorithm not only completely bypasses complex non-linear operations, guaranteeing the robustness and low Gas consumption of the contract, but at a microscopic scale, the error introduced by the second-order correction is strictly converged within a minimal range, ensuring the absolute fairness and rigor of the protocol's economic model.

3.4 Settlement Timing

To ensure logical causality, address potential invisible debt issues caused by long-term absence, and prevent time attacks, this protocol strictly splits the score settlement into two independent time checkpoints: when a challenge starts, and when a challenge settles.

3.4.1 When a Challenge Starts

When a user starts a new challenge, the contract immediately triggers a "Gap Entropy Settlement" and calculates the credit score change. This step aims to settle the user's time debt from the end of their last active period to the present, which can be simply understood as "settling old accounts".



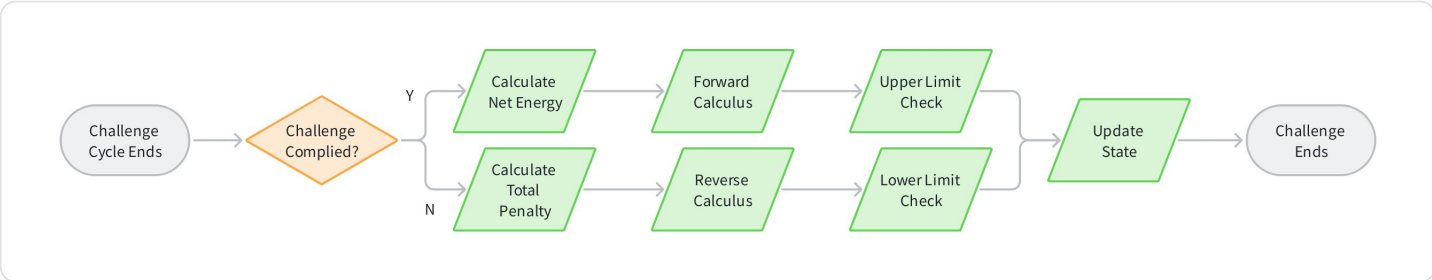
Flowchart of Gap Entropy Settlement upon Challenge Start

1. **Calculate Gap Entropy:**
   a. Read the last settlement time t_{last} and the current time t_{now} .
   b. Calculate the gap rest days: $g = (t_{now} - t_{last})$.
   c. Calculate the gap entropy debt: $E_{gap} = g \times E_{de}$.
2. **Execute Deduction and Exemption:**
   a. Execute reverse calculus: $S_{new} = S_t - \text{Integrate}(E_{gap})$.
   b. Survival Exemption: If the calculated result S_{new} touches the bottom line of 35.0, the descent is forcibly stopped, and the remaining undeducted entropy debt is automatically exempted by the system.
3. **Update On-chain State:** $S_t = S_{new}$, $t_{last} = t_{now}$.

The logical significance of step 2.b is: if a user undergoes a "bankruptcy reorganization," their previous debt of laziness is written off entirely. The user can start this challenge with a clean slate of 35.0, avoiding the negative experience of still paying off debts after training for a week.

3.4.2 When a Challenge Settles

When the challenge cycle ends, the contract calculates the credit score change based on the compliance results. This step focuses solely on the performance during the current challenge cycle, and can be simply understood as "settling new accounts".



Flowchart of Challenge End Settlement

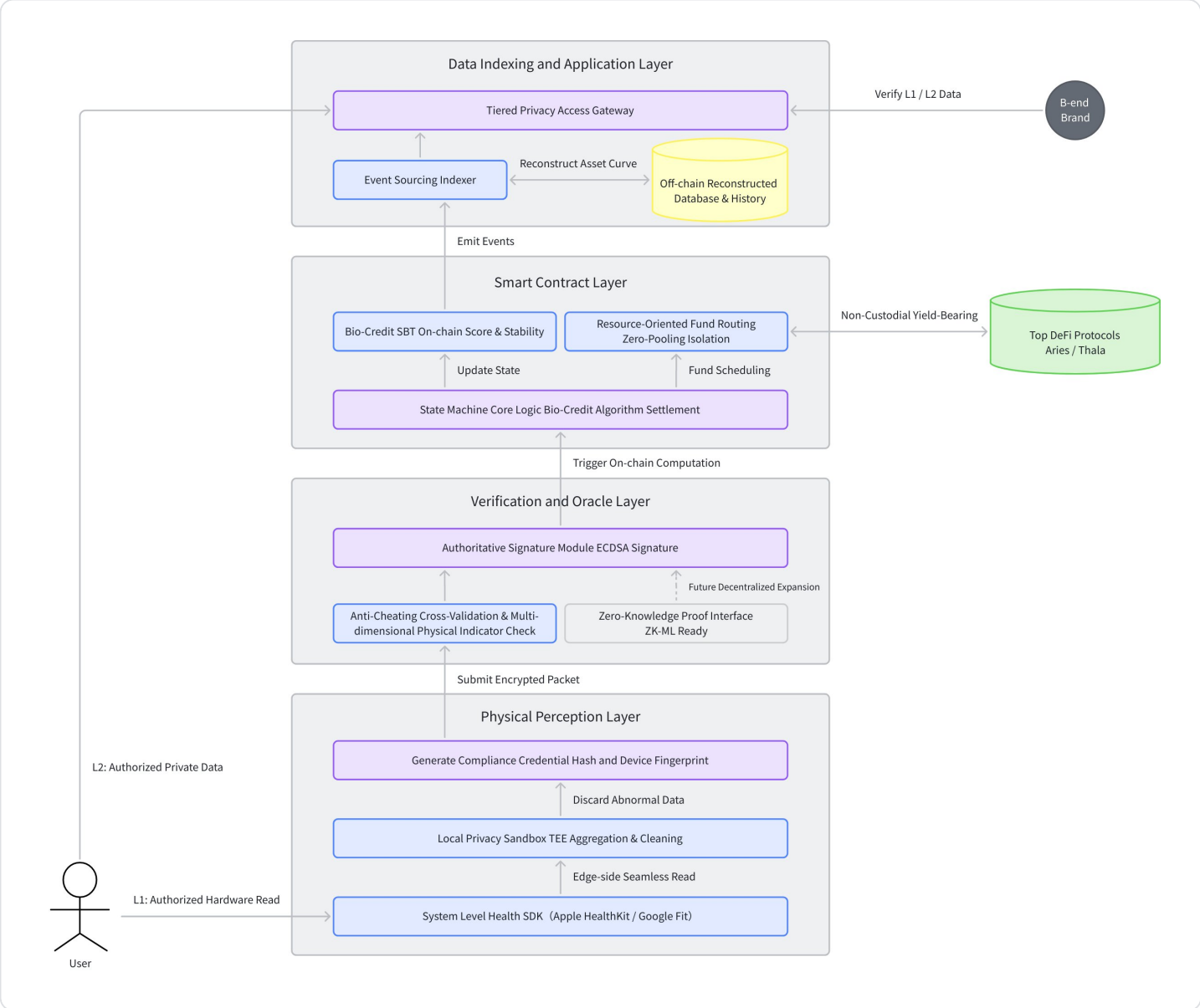
- **Branch A: Challenge Complied:**
  a. Calculate E_{net} using the logic provided in Section 2.2.2.
  b. Execute forward calculus (add points): $S_{new} = S_t + Integrate(E_{net})$.
  c. Boundary check to prevent exceeding the upper limit: $S_{new} = \min(S_{new}, 95.0)$.
- **Branch B: Challenge Defaulted:**
  a. Directly judge E_{net} as invalid.
  b. Calculate the energy to be penalized for the m days the target was not met: $E_{slash} = m \times E_{goal}$.
  c. Execute reverse integral (deduct points): $S_{new} = S_t - Integrate(E_{slash} + E_{de})$.
  d. Boundary check to prevent dropping below the lower limit: $S_{new} = \max(S_{new}, 35.0)$.
- **Update On-chain State:** $S_t = S_{new}$, $T_{last} = T_{now}$.

4 Technical Design, Efficiency, and Security

The philosophy of the architecture design is Layered Decoupling, Event-Driven, and a Strictly Non-custodial asset security model based on the Move language.

4.1 Overall Layered Architecture of the System

The system does not adopt the traditional Web2 Monolithic architecture where centralized servers handle everything. Instead, addressing the natural trust gap between off-chain physical behaviors and on-chain deterministic states, it has designed a rigorous four-layer architecture network. As shown in the figure below, from bottom to top, they are:



Overall Four-Layer Architecture of the System

4.1.1 Physical Perception Layer

Acting as the "tactile nerve" of the protocol to achieve an anti-cheating and seamless experience with zero manual input, this layer integrates the following automated pipelines internally and is deployed directly on the user's mobile devices or wearable hardware:

- **System-level SDK Direct Connection:** The client App embeds the native SDKs of Apple HealthKit (iOS) and Health Connect (Android). When a user registers with the protocol for the first time, the system will pop up an OS-level authorization prompt requesting read-only permissions for specific indicators such as "steps, active energy, sleep analysis." Once authorized, the App can directly initiate read-only queries to the underlying OS-level database via system APIs (e.g., HKSampleQuery in iOS) every time the user opens the app (or through background silent tasks), automatically fetching daily incremental data and completely eliminating any form of manual form input.

- **Data Traceability Tag Verification:** The OS health database contains not only numerical values but also data source device tags (e.g., whether the heart rate data was written by an Apple Watch or simulated by an unknown third-party pedometer App). When reading, the client executes strict whitelist filtering logic, only capturing data generated by highly trusted hardware (like native phone sensors, official smartwatches) and directly discarding data injected by unknown third-party applications at the device end.
- **Trusted Device Environment Proof:** To prevent hackers from using Android Emulators or Jailbroken/Rooted devices to forcefully tamper with the local database before submitting, the client App will call underlying hardware-level security services (such as Apple's DeviceCheck / App Attest API and Google's Play Integrity API) before packaging the cryptographic hash. The App must obtain a "device health proof" issued by the phone's physical security chip and combine it into a hash with the user's motion data packet. When the oracle verifies the signature off-chain, if an abnormal device environment is detected, the check-in will be directly rejected.

4.1.2 Verification and Oracle Layer

This layer is the "trust bridge" connecting the chaotic state of the physical world with the deterministic state of the blockchain:

- **Anti-Cheating Cross-Validation:** The Oracle (oracle node) receives the data packet from the perception layer and utilizes multidimensional cross-validation algorithms (e.g., detecting the physical correlation between step frequency, GPS displacement, and average heart rate) to filter out cheating behaviors such as pedometer simulation and multi-device trajectory overlap.
- **Authoritative Signature:** Once validation is passed, the oracle will use the protocol's authoritative private key to sign the compliance result via the Elliptic Curve Digital Signature Algorithm (ECDSA), generating an anti-forgeable "Compliance Credential."
- **Zero-Knowledge Proof Ready:** The current version's centralized signature verification is a highly efficient transitional solution for the early stage. This layer has reserved support for ZK-ML (Zero-Knowledge Machine Learning) in its interface design. In the future, cross-validation computations will be decentralized to the user's local device, and the oracle will only need to verify the ZK-Proof, thereby achieving complete decentralization and privacy isolation.

4.1.3 Smart Contract Layer

This is the "brain" and "clearing center" of the protocol, deployed on top of the high-performance Aptos public chain:

- **Core State Machine Execution:** As the decentralized brain of the protocol, the contract is responsible for receiving the anti-forgeable credentials passed by the oracle and strictly driving the calculus settlement logic described in Chapter II to complete the objective adjudication and state update of the Bio-Credit score.
- **Decentralized Asset Gateway:** The smart contract does not act as a centralized Liquidity Pool in the traditional sense, but rather as a secure routing hub. It is responsible for precisely connecting the user's staked funds to an external trusted ecosystem and executing automated liquidation and value redistribution at challenge settlement.

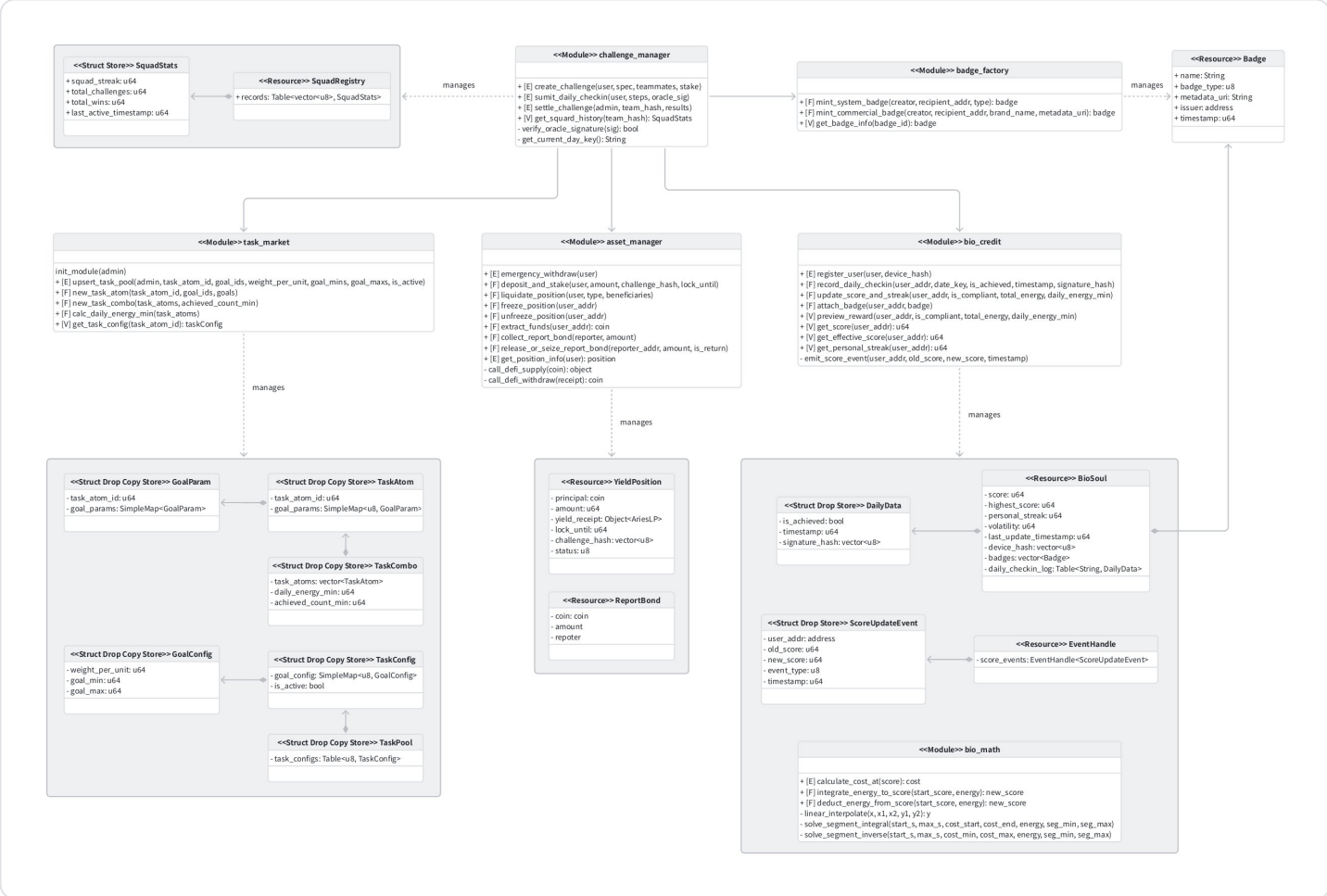
4.1.4 Data Indexing and Application Layer

This layer is responsible for translating the minimalist, low-cost underlying state on the public chain into high commercial value data dashboards readable by C-end users and B-end merchants:

- **Asynchronous Data Aggregation:** By deploying off-chain Indexers, it captures lightweight state change events thrown by the contract layer in real-time, converting obscure on-chain hexadecimal data into intuitive business charts, reproducing the complete historical trajectory without increasing the user's interactive Gas cost.
- **Tiered Privacy Gateway:** Provides standardized access channels for external DApps and B-end merchants. Combined with the signature authorization mechanism of Web3 wallets, it achieves strict tiered access to public reputation and personal behavioral privacy, completing the final commercial delivery where behavior equals value.

4.2 Smart Contract Module Design

The smart contract architecture of this protocol adopts a highly cohesive and loosely coupled modular design, as shown in the figure below. Based on the excellent security and resource-oriented characteristics of the Aptos Move language, we have divided the protocol into five core modules. This design not only enhances the auditability of the code but also reserves space for future functional expansions and upgrades.



Schematic of Smart Contract Modules

4.2.1 Challenge Manager

- This module is the master control node for business interactions, responsible for the scheduling and flow of the entire check-in challenge lifecycle.
- In the challenge initialization phase, it is responsible for generating a distributed hash teaming setup for regular users based on each team member's account address and the ID of the currently challenged task combo.
- For B-end merchants, it supports initiating whitelist mode challenges with no limit on the number of participants.
- At the end of the challenge cycle, it is responsible for triggering the final liquidation logic. Based on different judgment results such as full compliance, partial default, or full default circuit breaker, it schedules the Asset Manager to execute fund redistribution or complete confiscation.

4.2.2 Task Market

- This module acts as the protocol's behavioral definition center, managing all supported self-discipline indicators and rule parameters.
- It defines and maintains a set of standardized "Task Atoms" (e.g., daily exercise, compliant sleep, mindfulness meditation).

- It allows users to freely set parameters and synthesize multiple Task Atoms into independent "Task Combos".
- This module has built-in algorithms capable of automatically synthesizing the corresponding difficulty coefficient based on the parameter intensity of the task combo set by the user. It calculates the daily minimum energy limit, serving as the foundational weight for subsequent credit score calculations.

4.3.3 Asset Manager

- This module is the protocol's "treasury router," strictly executing a completely non-custodial fund scheduling logic.
- During the challenge's active period, it automatically routes users' staked funds to audited underlying yield-bearing protocols to obtain risk-free financial yields.
- It utilizes Move language features to achieve "Zero-Pooling" isolation. This ensures that the exchanged yield-bearing certificates (LP Tokens) are directly transmitted back and physically locked within the user's independent account resource space, thereby eliminating the risk of centralized malicious acts.
- It has a built-in "escape pod mechanism," which opens a mandatory withdrawal interface when the external oracle fails to respond overdue, allowing users to unconditionally retrieve their principal to eliminate the risk of a single point of failure.

4.3.4 Bio_Credit & Bio_Math

These two modules jointly constitute the protocol's core state machine and computational brain, responsible for processing the transformation of physical behaviors into on-chain reputation.

- **Bio_Credit**: Responsible for minting the non-transferable Bio-Credit SBT when a user first accesses the protocol, and forcibly requiring a unique hash mapping to be established with the user's physical health account. It also receives daily check-in results, manages energy states, and records core reputation data such as the user's consecutive compliance days.
- **Bio_Math**: Adopts scaled integers for underlying calculations to save Gas costs. It implements the protocol's original three-phase (Repair Phase, Accumulation Phase, Elite Phase) linear resistance model, as well as the real-time differential and integral algorithm for handling energy system entropy, ensuring that the growth of credit scores complies with objective biological laws.

4.3.5 Badge Factory

- This module is the protocol's incentive and commercial expansion gateway, responsible for the materialization of on-chain achievements.

- It is responsible for minting various achievement identifiers (such as winning streak badges) and commercial medals issued by the B-end, and mounting them as sub-resources under the user's Bio-Credit SBT.
- It grants credentials to users who reach a specific credit threshold (e.g., above 75 points), granting them whitelist permissions to participate in B-end merchant-funded tasks, thereby opening a direct conversion channel between self-discipline behaviors and commercial returns.

4.3 Storage Optimization and Event Sourcing

On high-performance public chains (such as Aptos), on-chain state storage is one of the most expensive computational resources. Traditional decentralized applications often fall into the trap of "state bloat"—piling up users' every check-in record and historical score change in the form of arrays (Vectors) within the smart contract, causing users' interactive Gas costs to rise exponentially over time.

Addressing this pain point, the protocol adopts an architectural paradigm of "minimalist on-chain state, off-chain event sourcing," ensuring the decentralization and tamper-proof characteristics of the data while achieving zero-cost historical queries.

4.3.1 Minimalist On-Chain State Storage

The protocol's smart contract abandons any form of array structure in the Global State. A user's BioCredit object permanently maintains only a snapshot of the current latest state on-chain (containing only the latest score, highest score, winning streak count, volatility, etc.), with complexity strictly controlled at $O(1)$. This means that no matter how long a user has been active in the protocol, the Gas fees consumed by their daily check-ins and settlements will always remain at an extremely low, constant baseline.

4.3.2 Event Sourcing

To meet the data visualization needs of C-end users (such as line charts of self-discipline credit score changes) and the historical risk control needs of B-end merchants, the protocol utilizes the Event mechanism of the Move language. For every calculus settlement of the credit score (whether adding or deducting points), the contract does not write the details into storage; instead, it emits a lightweight incremental event to the blockchain. This event contains the account address, scores before and after the change, the reason for the change (compliance/default), and the timestamp, etc.

4.3.3 Chain Indexing Reconstruction and Off-Chain Risk Control Database

- **Indexing**: The protocol deploys highly available Indexers (indexer nodes) off-chain to monitor and parse events on the Aptos chain in real-time.

- **Data Reconstruction**: By replaying these incremental events in chronological order, traditional off-chain structured databases (such as PostgreSQL) can replicate the complete historical curve and volatility trajectory of every user at an extremely low cost.
- **Verifiability**: Because the original events are permanently engraved in the public chain blocks, the off-chain reconstructed data possesses absolute verifiability. Anyone can safely and quickly fetch this low-cost aggregated data through the gateway provided by the protocol, without needing to initiate high-frequency query requests directly to expensive public chain nodes.
- **Tiered Invocation**: L1 public data (current credit score) is permissionlessly open; L2 private data (such as historical volatility curves, behavioral preferences) must be proactively signed and authorized by the user via their wallet on the frontend before merchants can fetch it through the gateway.

4.4 Identity Credibility and Sybil Resistance Security

In Web3 systems, security does not merely mean that smart contracts have no vulnerabilities; it also means that data and identities within the ecosystem are not maliciously manipulated. This protocol adopts a two-pronged approach of "physical anchoring" and "economic friction" to block Sybil Attacks and the speculative behavior of batch account farming from the root.

4.4.1 Physical-Level Uniqueness Anchoring

The protocol strictly requires a one-to-one hash mapping binding between the Bio-Credit SBT and the user's physical health account (such as Apple Health ID or Google Fit). By verifying the device health proof issued by the underlying operating system, the system directly rejects data injection from emulators or tampered devices, ensuring "one real person, one on-chain account."

4.4.2 Economic Friction Against Account Farming

Users need to pay a 0.75 USD protocol entry fee when creating an account for the first time. This seemingly minor threshold, combined with the binding restrictions of physical hardware, causes the cost for hackers to set up tens of thousands of fake nodes for "batch account farming" and "farming and restarting" to rise exponentially. While ensuring an extremely low threshold for genuine users, the system has constructed a solid anti-farming wall.

4.5 Non-Custodial Fund Routing and Ultimate Security

History in the Web3 domain has repeatedly proven: any centralized Liquidity Pool that aggregates massive amounts of user funds may ultimately be reduced to a "honeypot" for hackers, and even provides the physical conditions for a Rug Pull (absconding with funds) by the project team.

This protocol, however, based on the natively original Resource-Oriented feature of the Aptos Move language, has achieved an underlying asset management architecture of Zero-Pooling.

4.5.1 Completely Eradicating Centralized Honeypots

In the architecture of this protocol, there is no overarching protocol treasury. When a user stakes stablecoins (USDT/USDC) to participate in a challenge, these funds will not be transferred into a central contract address of this protocol. On the contrary, the funds always flow in isolation between various state machines in the form of independent Objects or Resources.

4.5.2 Lossless Routing and Flow of Yield-Bearing Funds

To ensure that users' staked assets can acquire Real Yield from DeFi during the challenge period, the protocol acts as a pure routing gateway (Router):

- **Automatic Deposit:** When a challenge becomes active, the contract will automatically call the standardized cross-chain/deposit interfaces of top-audited underlying DeFi protocols (such as Aries Market, Thala) on the Aptos chain to deposit the user's stablecoins and convert them into yield-bearing certificates (LP Tokens).
- **Physical Isolation of Certificates:** This is the highest barrier in the protocol's security design. The exchanged LP Token resources will be directly transmitted back and locked within the user's own Account Resource space. The protocol's contract code only holds the liquidation right and unlocking right of this asset through the capability model of the Move language.

4.5.3 Anti-Hacker and Anti-Malice Mechanisms

This architecture, where assets are retained in the user's physical account and permissions are scheduled by protocol rules, fundamentally performs a dimensionality reduction strike against various security threats:

- **Anti-Hacker Attack:** Even if the smart contracts of this protocol have unprecedented logical vulnerabilities, hackers cannot execute an operation to drain the liquidity pool with one click. This is because, on a physical level, the funds of 100,000 users are dispersedly stored in 100,000 independent account spaces; the attack surface for hackers is extremely narrow, and the cost is infinitely magnified.
- **Anti-Protocol Malice:** Because the core logic is open-source and immutable, apart from earning the profits declared by the protocol, the project team has no backdoor permissions to forcibly transfer the LP Tokens belonging to users, achieving "Code is Law" in its truest sense.

5 Protocol Philosophy and Governance Evolution

This protocol is not only a technical protocol, but also a sociological experiment on human self-discipline, attention reconstruction, and decentralized reputation. This chapter will elaborate on the core values driving the protocol's operation, as well as our evolutionary roadmap before reaching the endgame of complete decentralization.

5.1 Bayesian Credit

Originating from Bayes' Theorem in probability theory, which dynamically updates the probability of a hypothesis based on newly collected evidence. In the self-discipline credit philosophy of this protocol, the score accumulated by a user historically can be viewed as the "Prior Probability" of their self-discipline capability; meanwhile, the user's actual daily check-in result (compliance or default) serves as the new "Evidence". The protocol's core state machine is essentially executing a physical behavior-based Bayesian update, continuously calibrating and outputting a "Posterior Reputation" that best represents the user's current true willpower.

5.2 Credit Values: Pure Proof of Behavior

The protocol's judgment of self-discipline credit follows a cold but absolutely fair iron law: only the sweat of keeping promises is an asset; the sweat of breaking promises is merely in vain.

If a user is absent for most of the challenge period but frantically exercises on the last day to make up the total energy, this behavior will be ruthlessly judged as a default and penalized with point deductions by the protocol. If you think this is unfair, imagine a child who doesn't do homework during the holidays and crams on the last day; although the volume of homework is completed, they have actually lost the ability to plan and the quality of perseverance. If the protocol still awarded them points (or even just didn't deduct points), it would somewhat imply tacit approval or even encouragement of such speculative behavior.

This sends an extremely hardcore signal to all users: the protocol does not look at the hard work of trying, but only at the merit of compliance. This mechanism completely eradicates the "last-day crammers". This is exactly where the gold standard of Bio-Credit lies—when B-end brands see a high-credit user with 80 points on the network, they can be 100% certain that every single point of this score was earned through genuine, day-by-day perseverance, rather than farmed through a few bursts of whim.

5.3 Entry Friction and Web3 Enlightenment

The entry threshold of 0.75 USD set by the protocol, aside from serving as an anti-Sybil security mechanism, carries the enlightenment mission of Web3 breaking into the mainstream. Because the penetration rate of crypto wallets among mainstream populations is still relatively low, paying this minor fee conveniently forces the user to go through the entire process of "purchasing stablecoins $\rightarrow$ withdrawing to an on-chain wallet $\rightarrow$ signing the payment transaction". This greatly lowers the users' cognitive barrier towards decentralized applications,

accumulating valuable first-time on-chain interaction experience for their subsequent reception of financial yields and commercial incentives.

5.4 The Anti-Ponzi and Anti-Gambling Economic Undertone

In the design of its economic model, this protocol strictly abides by the absolute baseline of being "Anti-Ponzi". Unlike many Ponzi models that rely on the funds of late entrants to pay the interest of earlier participants, this protocol is a transparent and self-consistent closed system regarding its funding sources. All value flows within the protocol originate solely from three authentic channels:

1. The redistribution of defaulters' penalties.
2. The financial interest from underlying assets.
3. And the commercial capital injection from B-end brands.

Simultaneously, we resolutely oppose the alienation of self-discipline behaviors into high-leverage gambling. Through the supervisory mechanism of strong social relationships (forming groups of 4~7 acquaintances) and strict single-staking limits, the protocol rigorously restricts economic penalties to a reasonable range that aids self-discipline, completely stripping away any speculative and gambling attributes.

5.5 Progressive Decentralized Governance

To guarantee absolute security in the early stages, the initial parameters and update permissions for "Task Atom" objects are temporarily managed by the deployer, and we will be extremely cautious with any modifications. However, with the accumulation of high-credit users, the protocol will inevitably iterate rapidly (such as adding a large number of new task types, adjusting difficulty weights, etc.). Yet, extensibility may introduce the destruction of balance; any changes should be prudent and implemented only after being passed by community voting.

We envision a DAO governance model based on Bio-Credit. However, unlike traditional DAOs, the voting power that determines the protocol's expansion direction will no longer be simply determined by the amount of funds, but deeply bound to the user's Bio-Credit score. We firmly believe: only those who have demonstrated extremely high self-discipline and long-termism are most qualified to decide the future direction of this protocol.

5.6 Ultimate Vision: Absolute Privacy and Trustless Verification

The protocol currently utilizes a centralized Oracle to execute anti-cheating cross-validation. Although this reduces the degree of decentralization of the protocol, it is nevertheless a reasonable transitional approach for the early stages. Imagine, if anyone could deploy an open-source oracle equipped with anti-cheating verification algorithms, how to ensure that the oracle

network itself and its nodes do not cheat would become a new proposition. This might require designing another tokenomics and incentive mechanism, but this would deviate the focus of early development and increase difficulty.

Our ultimate vision is to achieve complete decentralization and privacy protection. The protocol has already reserved interfaces for Zero-Knowledge Machine Learning (ZK-ML) at the architectural layer. In the future, as edge-side computing power improves and ZKP algorithms mature, anti-cheating cross-validation will be decentralized to execute on the user's local mobile device. User data will permanently remain in local sandboxes, and the chain will only verify whether the zero-knowledge proof passes or fails. By then, we will have completed the most ambitious puzzle piece in the Web3 domain—proving the victory of human willpower with absolute mathematical certainty, without prying into any personal privacy.

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Appendix A: Glossary

- **SBT (Soulbound Token)**: A non-transferable, non-tradable digital asset used to construct a permanent decentralized identity and reputation on-chain. In this protocol, it is the sole carrier of Bio-Credit.
- **Task Atom**: The smallest, most fundamental verifiable unit of self-discipline behavior supported by the protocol (e.g., a single run, a single meditation session), which users can freely assemble into "Task Combos".
- **System Entropy**: Borrowed from the concept of thermodynamics, it refers to the minimum basal metabolic energy loss required to maintain a state of self-discipline. In the protocol, it manifests as a basic energy value automatically deducted daily.
- **Zero-Pooling (Zero-Pooling Isolation)**: The original asset management paradigm of this protocol. Users' staked funds are not aggregated in any centralized smart contract pool, but are physically isolated in their respective on-chain account spaces in the form of independent resources.
- **ZK-ML (Zero-Knowledge Machine Learning)**: A cryptographic technique that allows machine learning risk control models to run on local devices and prove the correctness of their results to the blockchain, without leaking any original input data (such as heart rate, step frequency) on-chain.

- **Stablecoin (e.g., USDT, USDC)**: Crypto assets whose value is pegged to fiat currencies like the US dollar. This protocol exclusively uses them as the sole staking and settlement standard across the board to eliminate the interference of crypto asset price volatility on behavioral incentives.
- **Ponzi Scheme / Anti-Ponzi**: A Ponzi scheme is an unsustainable financial game that relies on funds from late entrants to pay interest to early participants. This protocol strictly adheres to the "Anti-Ponzi" principle, ensuring that value flows within the protocol originate solely from authentic default redistributions, underlying DeFi financial interest, and B-end commercial capital injections.
- **Sybil Attack**: An attack in a peer-to-peer network where a single malicious entity creates and controls a large number of fake identities (nodes) to gain unfair advantages. This protocol blocks such batch account farming behaviors from the root through a 0.75 USD economic friction and physical device binding.
- **Move Language / Resource-Oriented**: A programming language specifically designed for secure smart contracts. Its "Resource-Oriented" feature defines digital assets as special variables that cannot be copied or implicitly discarded, greatly enhancing the security of non-custodial asset management.
- **Oracle**: A trust bridge connecting blockchain smart contracts with external real-world data. In this protocol, the oracle is equipped with anti-cheating algorithms, responsible for fetching and verifying users' physical hardware data, and generating anti-forgeable compliance credentials on-chain.
- **Gas (On-chain Computation/Interaction Fee)**: The cost required to execute transactions or smart contract state changes on a public chain network. Through clever mathematical algorithms (such as scaled integer calculus) and minimalist state storage, this protocol keeps users' Gas costs at an extremely low level.
- **Rug Pull**: A crypto industry term referring to the fraudulent behavior where a project team maliciously transfers and steals users' staked assets by exploiting contract backdoors or centralized liquidity pool permissions. The Zero-Pooling architecture of this protocol eliminates this risk.
- **RWE (Real World Evidence)**: Verified behavioral data originating from the real physical world. Under the premise of user authorization, the compliance data of high-credit users in this protocol can serve as RWE, providing a high-value data source for AI training and insurance actuarial science.
- **Real Yield**: Unlike illusory models that pay high interest by indiscriminately issuing platform tokens, Real Yield refers to real, principal-protected, and sustainable cash flow returns originating from actual economic activities (such as decentralized financial interest).

Appendix B: Core Parameter Derivation

The numerical constants of this protocol are not set randomly, but are derived through reverse engineering based on a "Standard User Model". Specifically, for a medium-intensity, single-task user (defined as completing only the "Daily Exercise" task daily, with an intensity of 500kcal + 60min), the theoretical cycle to transition from the Accumulation Phase (50 points) to the Elite Phase (75 points) is designed to be 168 days (considering real-life friction, practically about 180 days).

B.1 Basic Definitions

Target Milestone (ΔS): The score span from the Accumulation Phase (50.0) to the Elite Phase (75.0), noting the precision:

$$\Delta S = 25.0 \text{ Points} \times 10^6$$

Average Resistance ($\overline{C}$): The average energy cost within the 50~75 point interval, noting the precision:

$$\overline{C} = \frac{C_{start} + C_{end}}{2} = \frac{1.25 + 3.0}{2} = 2.125 \times 10^6$$

Target Period (T_{days}): The designed expected time to reach the goal (about 6 months):

$$T_{days} \approx 180 \text{ Days}$$

Standard Daily Difficulty (D_{std}): The base difficulty calculated based on daily exercise (500kcal + 60min):

$$D_{std} = (500 \times 1) + (60 \times 3) = 680$$

B.2 Energy Scaling Factor

The setting of the energy scaling factor (F_{scale}) aims to balance the total energy demand with the daily output capacity. The derivation formula is as follows:

$$F_{scale} \approx \frac{E_{net} + E_{de}}{D_{std}} = \frac{(\frac{\Delta S \cdot \overline{C}}{T_{days}}) + E_{de}}{D_{std}}$$

Substituting the values for estimation:

1. Required total net energy: $E_{total} = 25 \times 2.125 \times 10^6 = 53,125,000$.
2. Required daily net energy: $E_{daily_net} = 53,125,000/180 \approx 295,138$.
3. If the entropy value is set to (see B.3): $E_{de} = 25,000$.
4. Calculate the coefficient:

$$F_{scale} = \frac{295,138 + 25,000}{680} \approx 470.79$$

Final setting: To facilitate integer calculations and provide a slight acceleration incentive, it is rounded up to 500. After rounding, the theoretical cycle to reach the goal is shortened to $53,125,000/(680 \times 500 - 25,000) \approx 168.6$ days, which aligns with the design expectations.

B.3 Daily Energy System Entropy

The Daily Energy System Entropy (E_{de}) aims to simulate biological metabolic loss. Its value-setting principle is to cause a slight sense of pain, but not be fatal. Its percentage of the standard daily output energy is:

$$Ratio_{decay} = \frac{E_{de}}{E_{gross_std}} \times 100\%$$

Substituting the final parameters:

1. Standard daily output: $E_{gross_std} = 680 \times 500 = 340,000$。
2. If the entropy value is set to: $E_{entropy} = 25,000$。
3. Loss percentage:

$$Ratio_{decay} = \frac{25,000}{340,000} \times 100\% \approx 7.35\%$$

Conclusion: A natural decay rate of approximately 7% per day ensures the binding force of "sailing against the current," while allowing users to retain the vast majority of their accumulated achievements when sick or resting.